Isometric Standard Project Verification Report

For

Charm Industrial

Client	**Isometric**
Company / GHG Removal Supplier	**Charm Industrial**
Facility or Project Name	**Charm Kansas Bio-Oil Geologic Storage**
Facility Address	**Bio-oil Production:** **AE Cote-Nord Canada Bioenergy Inc (AECN)** Port Cartier, Quebec, Canada **Charm Miniforge** Fort Lupton, CO, USA **Bio-Oil Pre-Processing:** **Charm Industrial** El Dorado, KS, USA **Bio-oil Injection:** **Vaulted Deep** Hutchinson, KS, USA
Greenhouse Gas Statement ID	**Isometric ggs_1K32E9P3F1S0M8J6**
Net Greenhouse Gas Removals	**555.95 tonnes**
Removal Method	**Bio-Oil Geological Storage**
Removal Period	**June 1, 2025 – July 31, 2025**
Verification Opinion	**Positive (fully meets criteria)**
Verifier/Validation and Verification Body	**Tim Hansen** **350Solutions, Inc.**

ISSUED:

SEPTEMBER 3, 2025

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ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Guy Ingram Hardwick
Removal Method: Bio-Oil Geological Storage		
Report Date: September 3, 2025		
Document No: 350VR-Charm-ISO2504.05		
Rev: v1.2		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14 Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	June 1, 2025 – July 31, 2025
Greenhouse Gas Statement ID	Isometric ggs_1K32E9P3F1S0M8J6 (Statement ID) Isometric gss_1K32EB31K1S07XDV (Version ID)
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	555.95 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns

Verification Report Date	September 3, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of carbon removal credits by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.7 • Bio-Oil Geological Storage Protocol v1.1 • Biomass Feedstock Accounting Module v1.2 • Biomass or Bio-oil Storage in Salt Caverns v1.1 • Embodied Emissions Accounting v 1.0 • Energy Use Accounting v 1.2 • Transportation Emissions Accounting v1.1

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period including June 1, 2025 – July 31, 2025. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Kansas Bio-oil Geologic Storage project for use of bio-oil produced by Charm was completed on October 18, 2024, for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING JULY 31, 2025

Removal ID	Supplier reference ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1K2QHYCPB1S07P9F	863	2025-07-16	2025-07-16	46740
rmv_1K2QKKCR61S0D5XC	866	2025-07-21	2025-07-21	44070
rmv_1K2QHHD2T1S0XXM2	861	2025-07-14	2025-07-14	57440
rmv_1K2QM98DY1S06DAP	859	2025-07-10	2025-07-10	45530
rmv_1K2QMD2D31S099QB	860	2025-07-11	2025-07-11	44920
rmv_1K2QVFTBE1S0793M	867	2025-07-22	2025-07-22	42260
rmv_1K2NB4SE41S0CEYX	853	2025-07-01	2025-07-01	6560
rmv_1K2QJPRAH1S0RTMP	864	2025-07-17	2025-07-17	46600
rmv_1K2QVTKHJ1S0M9NR	868	2025-07-23	2025-07-23	45880
rmv_1K31PFEXB1S0M532	856	2025-07-07	2025-07-07	39120

rmv_1K2QGDTHD1S0121N	855	2025-07-02	2025-07-02	4660
rmv_1K2QHVT4H1S02CA4	862	2025-07-15	2025-07-15	2160
rmv_1K2NAMC2D1S0AN45	854	2025-06-26	2025-06-26	30470
rmv_1K2QKKE851S0JHV7	865	2025-07-18	2025-07-18	7650
rmv_1K2QH6B8V1S0G342	857	2025-07-08	2025-07-08	48070
rmv_1K2QM6RTX1S0Q5SP	858	2025-07-09	2025-07-09	43820
TOTAL				555950

The quantity of **555.95 tonnes** of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2.1*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VR-Charm-ISO2504.05

Timothy A. Hansen Digitally signed by Timothy A. Hansen Date: 2025.09.03 19:51:56 -05'00'

Signed: Tim Hansen (Verifier)

Guy Hardwick Digitally signed by Guy Hardwick Date: 2025.09.04 10:14:37 +01'00'

Guy Ingram-Hardwick (QA)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the referenced Reporting Period. The verification was conducted following the specifications of *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*. The verification is based on the provision and fair presentation of a complete and accurate GHG Statement submitted by Charm Industrial to the Isometric registry. This is confirmed and verified by 350Solutions based on supporting evidence in accordance with the Isometric criteria (Table 1, Section 3.2).

2.2. LEVEL OF ASSURANCE & MATERIALITY

The verification was conducted with a reasonable level of assurance. Any discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.3. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020:2012 for completion of ISO 14034:2016 Technology Verifications. 350Solutions is headquartered in Raleigh, North Carolina, USA. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was provided QA by Guy Ingram-Hardwick, a Carbon Removal Verification Engineer, with experience developing MRV requirements for and verifying carbon removal technologies.

2.4. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil Geologic storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
August 15-31, 2025	Review of reporting period modifications and changes	- Review of GHG Statement and report and guidance from Charm	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx June_July 2025 Removals Isometric GHG Statement Report v1.6.docx (1) VVB Guidance.docx ggs_1K32E9P3F1S0M8J6 (Version ID gss_1K32EB31K1S07XDV)

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
August 27 – September 2, 2025	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations and changes in RP - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues and findings and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx June_July 2025 Removals Isometric GHG Statement Report v1.6.docx (1) VVB Guidance.docx ggs_1K32E9P3F1S0M8J6 (Version ID gss_1K32EB31K1S07XDV) Isometric Standard V1.7.2, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.2
September 1-3, 2025	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information - Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtained bio-oil from two sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. In addition, Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Bio-oil from both facilities is shipped by tanker truck, rail car, or tote tanks on flatbed to a Charm facility in El Dorado, Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep

injection facility. During this Reporting Period, Charm also began shipping additional products for injection, including the aqueous phase (wood vinegar), which was approved for injection by the permitting authority.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored aboveground at the Vaulted site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are provided in their own PDD, which has been previously validated [1] and remains valid for the reporting period.

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.7 [2]
- Bio-Oil Geological Storage Protocol v1.1 [3]
- Biomass Feedstock Accounting Module v1.2 [4]
- Biomass or Bio-oil Storage in Salt Caverns v1.1 [5]
- Embodied Emissions Accounting v 1.0 [6]
- Energy Use Accounting v 1.2 [7]
- Transportation Emissions Accounting v 1.1 [8]

Charm utilizes Book and Claim Units (BCUs) to substitute for some, or all, or project diesel fuel usage and purchases Renewable Energy Certificates (RECs) for some or all its electricity consumption.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into salt caverns. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply of wood waste to the biomass processing sites (AECN and Charm Fort Lupton), the conversion of the biomass to bio-oil via AECN's pyrolysis process and Charm's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made modifications to the process, data collection, and calculation procedures which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period. All were reviewed and deemed acceptable under the Isometric requirements

- Charm has obtained a new supply of biomass during the Reporting Period, which was used for conversion into bio-oil during one month of the period.
    - The new biomass is of the same characteristics as prior approved biomass and is provided by the prior approved supplier, Altitude Forestry.
    - The biomass is non-merchantable wood from forestry thinning and wildfire prevention operations. The new biomass is from a separate project operated by Altitude, which is a government funded wildfire prevention project.
    - The biomass qualifies as EC5 for direct market leakage, EC8 for indirect market leakage, EC14 for counterfactual, and C1 (counterfactual is=0).
- Charm has had multiple disposals of bio-oil waste that have occurred due to issues during transportation and other issues.
    - All waste disposals have been accounted for in net emissions calculations and production emissions associated with bio-oil production and transport also accounted for.
    - Some waste bio-oil was used as substitute asphalt binder replacing bitumen. Charm claims zero emissions associated with disposal in this case. All production and transport emissions are accounted for. No avoided emissions are claimed.
    - Charm disposed of other materials through either waste incineration or landfilling. All production and transport emissions associated with the disposed material are included in the net removal calculations. Emission factors for incineration and landfilling were developed by Charm.
- Charm is now shipping both the bio-oil fraction of its liquid products as well as the Quench Oil / Wood Vinegar fraction to its processing facility and for injection. The first shipments and injections of its aqueous phase product occurred late in this reporting period
- Charm modified the process for wetting biochar for shipment and computing the mass of biochar. Charm now utilizes sampling of biochar from supersacks and determination of moisture content for calculation of dry biochar mass for allocation of process emissions between biochar and bio-oil products.
- Charm modified the calculation and monitoring approach for residual oil in totes after transport and injection. Charm measured weights of residual oil in 59 totes to determine an average amount of residual oil. Charm then utilizes a conservative estimate of residual oil of 125% of this value in determination of oil losses for calculation of total oil produced and shipped.
- Note that Charm is continuing to transition additional parts of its reporting to the Isometric Certify platform. As this occurs, certain components may appear partially complete or incomplete in Certify. However, the data has been submitted separately and verified by 350 as compliant. For this reporting period, Monitoring and Feedstocks requirements are included in Certify, but entries appear incomplete. However, these items are fully addressed outside of Certify.

3.6. PROCESS & DATA IRREGULARITIES DURING REPORTING PERIOD

Several anomalies and irregularities occurred during the reporting period related to data collection and operations. All have been reviewed, and the approach Charm has taken for each accepted and determined to be compliant with protocols. The following issues were identified by Charm and reviewed by 350Solutions:

- For one salt delivery to the El Dorado facility, delivery weigh scale tickets or bills of lading showing delivered salt weights were not provided by drivers. Charm used conservative approaches to account for delivery weights and the associated transportation emissions. Actual salt usage is documented clearly in operating logs.
- A propane delivery was completed in late May, but data not provided until beginning of this reporting period, so that information was included in this period.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil (oil and aqueous phases) by Charm during the period June 1, 2025 to July 31, 2025. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period, but properly accounted for and allocated to the bio-oil injected within the reporting period.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024, and continues for five years, ending February 28, 2028. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Review and confirmation of calculations of total bio-oil and CO_2 equivalent injected and all process, operational, and embodied emissions, as well as all supporting evidence for the following removals

rmv_1K2QGDTHD1S0121N	02 Jul 2025
rmv_1K2NB4SE41S0CEYX	01 Jul 2025
rmv_1K2QVTKHJ1S0M9NR	23 Jul 2025

- This includes for these removals:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS))

  - Review and calculation of average carbon content from raw data (lab reports)
  - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS
  - Confirmation of weigh scale raw data for deliveries of bio-oil and injectate
  - Re-calculation of transportation associated emissions for such transport
  - Confirmation of raw data for fuel usage at Charm pre-treatment facility
  - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
  - Confirmation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage emissions, respectively
  - Review of Charm pyrolysis process emissions, supporting evidence, and calculations
- Full Review of new waste disposal calculations and supporting data (Waste Disposal Emissions Calc Sheet.xls, June_July Railcar Cleaning Emissions Calc Sheet.xls and data)
- Review of new biomass pile shipment logs, biomass feedstock supply supporting data and review of biomass eligibility:
  - Altitude_Forestry_Supply_Agreement_-_Exhibit_A_(Arkansas_Mountain_Project)_(1).docx
  - Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL (1)
  - Evidence Package - Arkansas Mountain
  - Miniforge Biomass Transport Allocation -- Updated
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Meets the requirements with <u>minor revisions</u> of Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation. New feedstock properly accounted for with no leakage.
2.5.5	*Default Emission Factors, Proxies, and Models*	N	Acceptable – No additional significant changes were made since prior verifications. See Section 3.5 for summary of modifications to any calculation methods, all of which are acceptable. See Section 3.6 for description of irregularities, noting that all were properly and conservatively addressed in LCA net removal calculations.
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	
2.5.6	*Common Calculation Factors*	N	
2.5.7	*Uncertainty*	N	
2.5.8	*Durability and Monitoring*	Y	Storage Monitoring requirements were not fully met during Reporting Period. Isometric approved exception for this period only.
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	Y	Acceptable – No changes since prior verification. All documentation properly updated for reporting period
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV Certify platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete, and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Storage Monitoring requirements were not fully met during Reporting Period. Isometric approved exception for this period only. See Findings Table.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	Y	Acceptable – Additional feedstock utilized during period. All requirements met.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Integrity*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Acceptable - All required parameters measured and are within the required limits.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	NA	Acceptable - No change since validation
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	NA	Acceptable - No change since validation
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit, but reporting not completed on time. Isometric has approved an exception for report timing for this Reporting Period.
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1	*Permit for Operation*	N	Acceptable – No change since validation
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.1	*Sampling Frequency*	N	Acceptable – No change since validation. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

5.4.1. CRITICAL FINDINGS AND EXCEPTIONS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed. Findings of note include the following:

ID No	Type	Finding / Issue	Supplier Response	Conclusion / Resolution
11	Non-Conformity (Omission / Not material)	Please provide the bills of lading to validate the Mass shipped for the rail car waste disposal	The mass shipped for railcar waste disposal is documented by the BOLs that AECN provided for the truckloads that were transferred to the railcar. This is the most conservative estimate as it does not account for any loss of mass during the disposal process. This maximizes the estimated emissions as the emission factors for transportation and disposal are	Complete. Approach acceptable and conservative.

			scaled by mass. BOL numbers for each railcar are noted at the top of the calc sheet for each railcar. Please note, GATX 023958 and 023975 were never filled, so there are no BOLs or disposal emissions listed for those railcars. Bills of lading are provided in a separate folder that I will send over	
12	Non-Conformity (Omission / Not material)	Can you please point to where the embodied emissions associated with Vaulted and the Vaulted upgrades are included on Certify?	This appears to be a miss -- We updated the CI on the Ledger side to reflect the updated value, but there is no corresponding emission in Certify. After investigating the issue, this appears to be the case in general. We reconstructed the emissions impact by manually scaling the mass of injectate sequestered in each Reporting Period by the appropriate Vaulted embodied emissions CI. The result shows that this discrepancy is clearly sub-material; total Vaulted embodied emissions accounted for .008% of reporting period removals at the low end and .041% at the high end, totalling 482.7 kgs from project initiation in March of 2024 through 7/31/25. While this is clearly non-material, we do want to remedy it to ensure high quality reporting. We will update the next Reporting Period with a 482.7 kg site emission in order to net out all outstanding emissions. Moving forward, our transition to Mangrove/Certify will ensure this is not a repeat occurrence	Complete. To be reviewed at next verification. Approach is acceptable for addressing accurate reporting. Error is not material.
13	Non-Conformity (Misstatement / Not material)	On the Miniforge Biomass Transport Allocation Spreadsheet, the old emission factors appear to be used on the New Biomass Allocation Tracker tab. This only impacts the emissions by 0.04 tons total and is not material, but should be corrected.	Noted and corrected	Complete. Error was corrected and is not material.
14	Clarification	The tare weight used on the NEW Delivery Log tab refers to a tare weight for a 1995 Peterbilt (Tare Scale Ticket tab). When the cited delivery vehicle is a 2007 Peterbilt. This could have potential impact on emissions as the tare weights for prior delivery vehicles are on the order of 35k-36k lbs, while the tare weight for the new delivery vehicle is 45k lbs. Please clarify if this is correct and if a tare weight for the actual delivery vehicle is available.	This was a typo -- I have confirmed with our logistics team that the tare weight corresponds to the 2007 Peterbilt with Cummins motor and Reliance trailer, which is what was used for the deliveries. The emissions calculations are accurate, but the labeling of the tare weight is not and will be revised.	Complete. Correction acceptable. No change to verification data.

18	Nonconformity - Omission	The Reporting for the storage site is lacking for the reporting period. The following issues were identified: - the July Monthly Report has not been provided and no evidence is provided that it was submitted on time (August 28) - The Q2 Quarterly Report was not submitted on time (due July 28, but extension granted to Aug 28) and is not provided - The Q1 Quarterly Report was submitted late (April 2 instead of March 28) - A quarterly Groundwater Monitoring Report has not been provided for Q2 - The June Monthly Report was submitted late (an extension was granted) This reporting lapse causes significant concerns regarding permit compliance currently and in the future.	Noted	Complete. Isometric has approved a deviation for this reporting period only. Future periods may not proceed with verification finalization until all reports are provided. (Slack Message Tom Sellers August 15, 2025)
21	Clarification	The July 28 sample of displaced brine was much higher than the June and July 7 samples (3290 mg/l vs ~600). Is there any explanation or concerns regarding this?	Yes -- This was discussed with our principal scientist once received.	No action required. Monitor in future verifications.

An exception to protocol requirements was confirmed by Isometric regarding the late reporting as required by the storage location permit. The late reporting included the following during this Reporting Period:

- the July Monthly Report has not been provided, and no evidence is provided that it was submitted on time (August 28)
- the Q2 Quarterly Report was not submitted on time (due July 28, but extension granted to Aug 28) and is not provided
- the Q1 Quarterly Report was submitted late (April 2 instead of March 28
- the quarterly Groundwater Monitoring Report has not been provided for Q2
- The June Monthly Report was submitted late (an extension was granted)

According to Isometric (Slack Message Tom Sellers to Charm and 350, August 15, 2025): *"we'll use the following approach to reconciliation for this cycle:*

- *Monitoring reports (monthly + q2): can be submitted post-package this time. Missing reports without due notice in future may trigger a pause in verification.*
- *The missing data will need to be provided before the next verification. "*

5.4.2. FORWARD ACTION REQUESTS & RECOMMENDATIONS

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open Forward Action Requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

ID	Type	Issue	Supplier Response	Resolution
2	FAR	The PDD should be updated to reflect the new biomass source and associated information	The new biomass source is included in the PDD expansion currently under validation (see Appendix 3 Charm Biomass Feedstock Information -- Arkansas Mountain	Complete.
3	FAR	The PDD should be updated to reflect GHG Statement calculation changes as noted in the referenced briefs	Will update Section F of V1.10 with any changes not already captured	To be reviewed at next verification.
4	FAR	Please plan to provide updated documentation on community engagement and stakeholder input activities by 12/31/2025	Working with our Community Engagement Manager on this item	Review for future verifications
5	FAR	Please plan to provide updated documentation on grievance mechanism updates and any complaints/grievances received and responses (or confirmation of no issues) by 12/31/2025	Working with our Community Engagement Manager on this item	Review for future verifications
6	FAR	Please ensure all waste disposal related emissions for future disposal of wastes which have been produced during this period are accounted for in future reporting periods as required. Note: Charm has properly accounted for all disposals so far and has all systems in place to do so going forward.	We will continue to account for completed disposals as soon as they are completed and, if applicable, documentation is provided to quantify mass, confirm disposal method, etc.	Review for future verifications
19	FAR	For gas monitoring required by the permit, can you please provide a log of gas monitor results such that we can review and verify that no gas monitor levels have exceeded permit levels which require action? Please provide in next reporting period.	We will request these records from Vaulted	Review for future verifications
20	FAR	Please provide the following reports no later than the next reporting period end: - the July Monthly KOLAR Report - The Q2 Quarterly Report - The Q2 quarterly Groundwater Monitoring Report	We are currently waiting on Vaulted to provide these documents	To be reviewed at next verification

22	Forward Action Request (Prior)	350 is aware that Vaulted is completing some equipment and facility upgrades. If applicable to Charm injection processes, the embodied emissions should be included in Charm LCA.	Complete - Vaulted updates and equipment additions (May 2025) added to embodied emissions calculations in this reporting period	Complete.
23	Forward Action Request (Prior)	Non-pyrolysis electricity consumption (G&A, Eng) is based on records from May 2024. Please evaluate current usage for non-pyrolysis activity, as the changes in operations at the facility have been significant and usage may be much different. NEW Comment: Despite stating this would be addressed in this reporting period, the production period calculations still rely on the May 2024 allocation. Please ensure that this is addressed in the next reporting period.	Please complete and update by end of next reporting period	Complete. Explanation and update provided. Conservative approach used by using May 2024 data
24	Forward Action Request (Prior)	Lack of data obtained for salt deliveries as noted in Charms Verifier Guidance document provides less certainty in associated emissions and is an indicator of potential data tracking and management issues. Please work with suppliers to ensure proper weigh tickets / BOL are provided for each delivery to ensure accuracy. It is noted that the approach taken to fill data gaps appears conservative and overall does not materially effect the calculations of net removals. However, it does not instill confidence.	Charm has improved tracking significantly and was only missing one delivery ticket. Acceptable, but continue to monitor to ensure full implementation and consistency.	

6. REVISION HISTORY

Original date of issue: September 2, 2025

Version	Date Issued	Noted Changes
V1.0 (Draft – Internal)	September 2, 2025	N/A Initial Version
V1.1 (Draft – External)	September 2, 2025	Minor edits from QA
V1.2 (FINAL)	September 3, 2025	Final Version – Minor Edits from Isometric and Charm; Closure of Findings List

References

[1] "Vaulted Deep Great Plains Facility Organic Waste Sequestration: Isometric PDD." Jan. 23, 2025. Accessed: Feb. 19, 2025. [Online]. Available: https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H43SYY631S0Z8YR/prj_1HHYZFVGW1S044ZY/pjd_1JJSMS9FS1S0MF1M/pdd.pdf

[2] Isometric, "Isometric Standard v1.2.0." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[3] Isometric, "Bio-oil Geological Storage V1.1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[4] Isometric, "Biomass Feedstock Accounting V1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[5] Isometric, "Biomass or Bio-oil Storage in Salt Caverns V1.0.4." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[6] Isometric, "Embodied Emissions Accounting v1.0.3." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[7] Isometric, "Energy Use Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[8] Isometric, "Transportation Emissions Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

File Name	File Category
Altitude_Forestry_Supply_Agreement_-_Exhibit_A_(Arkansas_Mountain_Project)_(1).docx.pdf	Feedstock
Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL (1).pdf	Feedstock
Evidence Package - Arkansas Mountain.pdf	Feedstock
01 02 - 2025.04 Waste Determination Cover Letter, Process Description.docx	Aqueous Fraction Approvals
01 02 - 2025.04 Waste Determination Cover Letter, Process Description.pdf	Aqueous Fraction Approvals
03 - 2025.04.18 Waste Determination Form.pdf	Aqueous Fraction Approvals
04 - Wood Vinegar _ Quench Oil SDS.pdf	Aqueous Fraction Approvals
05 - TCLP.pdf	Aqueous Fraction Approvals
2025.04 Waste Determination for Fast Pyrolysis Bio-oil - signed.pdf	Aqueous Fraction Approvals
2025.04 Waste Determination for Fast Pyrolysis Bio-oil.pdf	Aqueous Fraction Approvals
APPROVED - 7.2.2025.png	Aqueous Fraction Approvals
Form O-175 Process Knowledge Permit Application - Charm Waste Profile 2025.04.pdf	Aqueous Fraction Approvals
Form O-450B Manifest - Process Knowledge Organics - Charm Waste Profile 2025.04 R2.pdf	Aqueous Fraction Approvals
June_July Removals Isometric GHG Statement Report v1.6.docx	\Additional Documentation\
VVB Guidance.docx	\Additional Documentation\
[Extenal] Bio-Char MRV Process Update.docx	\Additional Documentation\Process Change-Update Documents\
[External] Standardized Tote Loss Proposal v3.docx	\Additional Documentation\Process Change-Update Documents\
Charm BO Deliveries June+July.xlsx	\Charm BO MRV\Charm BO Deliveries\
Production Period 7 Summary.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\
PP7 (6_1-6_30) Biomass Throughput.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Throughput\
Allocation Calc Sheet.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\
Delivery Instructions.pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\
061825 - Tare Weight Scale Tickets (1).HEIC	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\
061825 - Tare Weight Scale Tickets.HEIC	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\
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250618-RW01.HEIC	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\

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250620-RW02.JPG	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\
250621-RW01.JPG	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\
250621-RW02.JPG	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Biomass Transport\New Log Pile\Scale Tickets\
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PP7 (6_1-6_30) Vendor Delivery Emissions Calc Sheet.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Consumables Delivery\
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6_25_25.PDF	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Diesel\Transaction Docs\
6_30_25.PDF	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Diesel\Transaction Docs\
6_3_25.PDF	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Diesel\Transaction Docs\
6_9_25.PDF	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Diesel\Transaction Docs\
Date Range Screenshot.png	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Electricity\
Power Data by Day.csv	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Electricity\
Power Data by Day.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Electricity\
PP7 Miniforge Electricity Consumption Calculator.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Electricity\
Cap Ex MRV June.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Embodied Emissions\
SP&C June.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Embodied Emissions\
PP7 Gasoline Consumption Calculator.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Gasoline\
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Production Period 7 Output Summary.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Output Data\
AQUEOUS 20250625 CAR-347.pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Output Data\Testing\
CHAR 20250701 CAR-351.pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Output Data\Testing\
WODO 20250630 CAR-350.pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Output Data\Testing\
Propane Consumption Calculator 6_1_25-6_30_25.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Propane\
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Manifest_HH245894463 (1).pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Waste Disposal\
PP7 Waste Disposal Emissions Calc Sheet.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Waste Disposal\
PP7 Water Usage Log.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 7\Water\

Production Period 8 Summary.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\
PP8 (7_1-7_31) Biomass Throughput.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Biomass Throughput\
Miniforge Biomass Transport Allocation -- Updated.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Biomass Transport\
PP8 (7_1-7_31) Vendor Delivery Emissions Calc Sheet.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Consumables Delivery\
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Production Period 8 Output Summary.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Output Data\
Biochar C Content Testing.pdf	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Output Data\Testing\
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Propane Consumption Calculator 7_1_25-7_31_25.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Propane\
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PP8 Water Usage Log.xlsx	\Charm BO MRV\Charm Production MRV Docs\Production Period 8\Water\
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Scale Tickets.jpg.pdf	\Lot Data\2-100\
Batch 2-100, 2-101-1 (CHN & H2O) (1).pdf	\Lot Data\2-100\Testing\
Batch 2-100, 2-101-1 (TAN & Density) (1).pdf	\Lot Data\2-100\Testing\
2-101 MRV Ops Notes.pdf	\Lot Data\2-101\
Scale Ticket 1-2.jpg.pdf	\Lot Data\2-101\

Scale Ticket 3.jpg.pdf	\Lot Data\2-101\
Batch 2-101-2 (CHN & H2O).pdf	\Lot Data\2-101\Testing\
Batch 2-101-2, 2-102 (TAN & Density).pdf	\Lot Data\2-101\Testing\
2-102 MRV Ops Notes.pdf	\Lot Data\2-102\
Scale Tickets.jpg.pdf	\Lot Data\2-102\
Batch 2-101-2, 2-102 (TAN & Density) (1).pdf	\Lot Data\2-102\Testing\
Batch 2-102 (CHN & H2O).pdf	\Lot Data\2-102\Testing\
2-99 Ops Notes.pdf	\Lot Data\2-99\
Scale Ticket 2-99.pdf	\Lot Data\2-99\
Batch 3-29 & 2-99 (CHN & H2O).pdf	\Lot Data\2-99\Testing\
Batch 3-29 & 2-99 (TAN & Density).pdf	\Lot Data\2-99\Testing\
3-29 Charm Contents.xlsx	\Lot Data\3-29\
3-29 MRV Ops Notes.pdf	\Lot Data\3-29\
Scale Ticket 1.jpg.pdf	\Lot Data\3-29\
Scale Ticket 2.jpg.pdf	\Lot Data\3-29\
Batch 3-29 & 2-99 (CHN & H2O).pdf	\Lot Data\3-29\Testing\
Batch 3-29 & 2-99 (TAN & Density).pdf	\Lot Data\3-29\Testing\
3-31 Charm Contents.xlsx	\Lot Data\3-31\
3-31 MRV Ops Notes.pdf	\Lot Data\3-31\
3-31 Scale Ticket 1.jpg.pdf	\Lot Data\3-31\
3-31 Scale Ticket 2.jpg.pdf	\Lot Data\3-31\
Batch 3-31 (CHN & H2O).pdf	\Lot Data\3-31\Testing\
Batch 3-31 (TAN & Density).pdf	\Lot Data\3-31\Testing\
3-32 Charm Contents.xlsx	\Lot Data\3-32\
3-32 MRV Ops Notes.pdf	\Lot Data\3-32\
3-32 Scale Tickets.jpg.pdf	\Lot Data\3-32\
Batch 3-32, 3-33 (CHN & H2O).pdf	\Lot Data\3-32\Testing\
Batch 3-32, 3-33 (TAN & Density).PDF	\Lot Data\3-32\Testing\
3-33 Charm Contents.xlsx	\Lot Data\3-33\
3-33 MRV Ops Notes.pdf	\Lot Data\3-33\
Scale Tickets.pdf	\Lot Data\3-33\
Batch 3-32, 3-33 (CHN & H2O).pdf	\Lot Data\3-33\Testing\
Batch 3-32, 3-33 (TAN & Density).PDF	\Lot Data\3-33\Testing\
3-34 Charm Contents.xlsx	\Lot Data\3-34\
3-34 MRV Ops Notes.pdf	\Lot Data\3-34\
3-34 Scale Ticket 2.jpg.pdf	\Lot Data\3-34\
3-34Scale Ticket 1.jpg.pdf	\Lot Data\3-34\
Batch 3-34 (CHN, H2O, TAN, Density).pdf	\Lot Data\3-34\Testing\
3-35 Charm Contents.xlsx	\Lot Data\3-35\
3-35 MRV Ops Notes.pdf	\Lot Data\3-35\
Scale Ticket 1 + 2.pdf	\Lot Data\3-35\

Batch 3-35 & 3-36 (CHN, H2O, TAN, Density).pdf	\Lot Data\3-35\Testing\
3-36 Charm Contents.xlsx	\Lot Data\3-36\
3-36 injection Scale ticket.jpg.pdf	\Lot Data\3-36\
3-36 MRV Ops Notes.pdf	\Lot Data\3-36\
Batch 3-35 & 3-36 (CHN, H2O, TAN, Density).pdf	\Lot Data\3-36\Testing\
3-37 Charm Contents.xlsx	\Lot Data\3-37\
3-37 MRV Ops Notes.pdf	\Lot Data\3-37\
Scale Tickets.pdf	\Lot Data\3-37\
BATCH 4-03 3-37 3-38 (CHN & H2O).pdf	\Lot Data\3-37\Testing\
BATCH 4-03 3-37 3-38 (TAN & Density).pdf	\Lot Data\3-37\Testing\
3-38 Charm Contents.xlsx	\Lot Data\3-38\
3-38 MRV Ops Notes.pdf	\Lot Data\3-38\
3-38 Scale Tickets.jpg.pdf	\Lot Data\3-38\
BATCH 4-03 3-37 3-38 (CHN & H2O).pdf	\Lot Data\3-38\Testing\
BATCH 4-03 3-37 3-38 (TAN & Density).pdf	\Lot Data\3-38\Testing\
3-39 Charm Contents.xlsx	\Lot Data\3-39\
3-39 MRV Ops Notes.pdf	\Lot Data\3-39\
3-39 Scale Tickets.pdf	\Lot Data\3-39\
Batch 3-39 (CHN & H2O).pdf	\Lot Data\3-39\Testing\
Batch 3-39 (TAN & Density).PDF	\Lot Data\3-39\Testing\
4-02 Charm Contents.xlsx	\Lot Data\4-02\
4-02 MRV Ops Notes.pdf	\Lot Data\4-02\
4-02 Scale Ticket 1.jpg.pdf	\Lot Data\4-02\
Batch 4-02 (CHN & H2O).pdf	\Lot Data\4-02\Testing\
Batch 4-02 (TAN & Density).pdf	\Lot Data\4-02\Testing\
4-03 Charm Contents.xlsx	\Lot Data\4-03\
4-03 MRV Ops Notes.pdf	\Lot Data\4-03\
4-03 Scale Ticket 1.png.pdf	\Lot Data\4-03\
4-03 Scale Ticket 2.png.pdf	\Lot Data\4-03\
BATCH 4-03 3-37 3-38 (CHN & H2O).pdf	\Lot Data\4-03\Testing\
BATCH 4-03 3-37 3-38 (TAN & Density).pdf	\Lot Data\4-03\Testing\
BOL 2169-T pickup 5.15 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2171-T pickup 5.18.pdf	\Lot Data\AECN BOL_s\
BOL 2172-T pickup 5.21 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2173-T pickup 5.21.pdf	\Lot Data\AECN BOL_s\
BOL 2174-T pickup 5.22.pdf	\Lot Data\AECN BOL_s\
BOL 2175-T 6.2.pdf	\Lot Data\AECN BOL_s\
BOL 2176-T 6.3.pdf	\Lot Data\AECN BOL_s\
BOL 2177-T 6.3.pdf	\Lot Data\AECN BOL_s\
BOL 2178-T 6.4.pdf	\Lot Data\AECN BOL_s\
BOL 2179-T 6.5.pdf	\Lot Data\AECN BOL_s\

BOL 2180-T 6.5 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2181-T 6.5 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2182-T 6.19.pdf	\Lot Data\AECN BOL_s\
BOL 2183-T 6.23 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2184-T 6.23.pdf	\Lot Data\AECN BOL_s\
BOL 2185-T 6.24.pdf	\Lot Data\AECN BOL_s\
BOL 2186 T 6.30.pdf	\Lot Data\AECN BOL_s\
BOL 2187-T 6.30 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2188-T pickup 7.2 (1).pdf	\Lot Data\AECN BOL_s\
BOL 2189-T pickup 7.3.pdf	\Lot Data\AECN BOL_s\
BOL 2190-T pickup 7.4.pdf	\Lot Data\AECN BOL_s\
BOL 2191-T pickup 7.5.pdf	\Lot Data\AECN BOL_s\
BOL 2192-T pickup 7.9.pdf	\Lot Data\AECN BOL_s\
BOL 2193-T Pickup 7.9.pdf	\Lot Data\AECN BOL_s\
BOL 2194-T pickup 7.10.pdf	\Lot Data\AECN BOL_s\
BOL 2195-T pickup 7.12.pdf	\Lot Data\AECN BOL_s\
7_10_25.pdf	\Site Emissions\Diesel\
7_30_25.pdf	\Site Emissions\Diesel\
7_31_25.pdf	\Site Emissions\Diesel\
June_July Generator Diesel Consumption Calcs.xlsx	\Site Emissions\Diesel\
Charm Industrial INC_csv_2025-08-04_20_00_19.csv	\Site Emissions\Electricity\Minifinery\
Charm Industrial INC_csv_2025-08-04_20_00_19.xlsx	\Site Emissions\Electricity\Minifinery\
Evergy Bill Screenshot.jpg	\Site Emissions\Electricity\Minifinery\
Minifinery June_July Consolidated Energy Consumption.xlsx	\Site Emissions\Electricity\Minifinery\
Vaulted Consolidated Electricity Consumption June-July.xlsx	\Site Emissions\Electricity\Vaulted\
185-11422.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11437.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11498.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11501.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11520.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11526.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11549.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11607.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11608.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11633.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11634.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11648.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11649.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11671.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11672.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11689.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\

185-11691.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11703.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11704.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11712.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11718.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11720.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11740.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11741 (1).pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11774.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11775.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11779.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11781.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11800.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11801.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11828.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-11829.pdf	\Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
June_July Gasoline Tracker KS.xlsx	\Site Emissions\Gasoline\
6_11_25.jpg	\Site Emissions\Gasoline\Receipts\
6_18_25.jpg	\Site Emissions\Gasoline\Receipts\
6_24_25.jpg	\Site Emissions\Gasoline\Receipts\
6_25_25.jpg	\Site Emissions\Gasoline\Receipts\
6_2_25(1).jpg	\Site Emissions\Gasoline\Receipts\
6_2_25.jpg	\Site Emissions\Gasoline\Receipts\
6_9_25.jpg	\Site Emissions\Gasoline\Receipts\
7_15_25.jpg	\Site Emissions\Gasoline\Receipts\
7_16_25.jpg	\Site Emissions\Gasoline\Receipts\
7_22_25.jpg	\Site Emissions\Gasoline\Receipts\
7_27_25.jpg	\Site Emissions\Gasoline\Receipts\
7_8_25.jpg	\Site Emissions\Gasoline\Receipts\
6_5_25.PDF	\Site Emissions\LCS\
7_10_25.PDF	\Site Emissions\LCS\
7_21_25.jpeg	\Site Emissions\LCS\
7_22_25.PDF	\Site Emissions\LCS\
7_3_25.PDF	\Site Emissions\LCS\
June_July LCS Consumption Sheet.xlsx	\Site Emissions\LCS\
Charm Salt Delivery Calc Sheet June_July.xlsx	\Site Emissions\Salt Deliveries\
IMG_3115.jpg	\Site Emissions\Salt Deliveries\
IMG_3116.jpg	\Site Emissions\Salt Deliveries\
IMG_3117.jpg	\Site Emissions\Salt Deliveries\
IMG_3118.jpg	\Site Emissions\Salt Deliveries\
IMG_3119.jpg	\Site Emissions\Salt Deliveries\

IMG_8165 (1).jpg	\Site Emissions\Salt Deliveries\
IMG_8167 (1).jpg	\Site Emissions\Salt Deliveries\
Scale Ticket 1.pdf	\Site Emissions\Salt Deliveries\
Scale Ticket 2.pdf	\Site Emissions\Salt Deliveries\
Scale Ticket 3.pdf	\Site Emissions\Salt Deliveries\
Scale Ticket 4.pdf	\Site Emissions\Salt Deliveries\
Scale Ticket 5.pdf	\Site Emissions\Salt Deliveries\
Scale Ticket 6.pdf	\Site Emissions\Salt Deliveries\
El Dorado ---_ Red Carpet Distance.png	\Site Emissions\Waste Disposal\
El Dorado--_Wichita--_Tulsa.png	\Site Emissions\Waste Disposal\
Waste Disposal Emissions Calc Sheet.xlsx	\Site Emissions\Waste Disposal\
June_July Railcar Cleaning Emissions Calc Sheet.xlsx	\Site Emissions\Waste Disposal\Railcar Disposals\
Copy of AECN to SOPOR (Baie Comeau).png	\Site Emissions\Waste Disposal\Railcar Disposals\Distance Measurements\
Copy of SOPOR to Maize (rail).jpeg	\Site Emissions\Waste Disposal\Railcar Disposals\Distance Measurements\
Maize to Ardmore + Ardmore to Sterling.png	\Site Emissions\Waste Disposal\Railcar Disposals\Distance Measurements\
SOPOR to Clean Harbors CN.png	\Site Emissions\Waste Disposal\Railcar Disposals\Distance Measurements\
SOPOR to TBC.png	\Site Emissions\Waste Disposal\Railcar Disposals\Distance Measurements\
6_11_25 Cleanout.pdf	\Site Emissions\Waste Disposal\Tank Cleanouts\
7_24_25 Cleanout.pdf	\Site Emissions\Waste Disposal\Tank Cleanouts\
Tank Cleanouts Calc Sheet June_July.xlsx	\Site Emissions\Waste Disposal\Tank Cleanouts\
6-25-25 AR-09 Displaced Brine.pdf	\Vaulted MRV\Brine TOC Reports\
AR-09 Displaced Brine 7-28-25.pdf	\Vaulted MRV\Brine TOC Reports\
AR-09 Displaced Brine 7-7-2025.pdf	\Vaulted MRV\Brine TOC Reports\
2025 Facility Proposed Expansion and Modification Plan 4.8.25.pdf	\Vaulted MRV\Facility expansion calculation and evidence\
2025-04-22_Cavern Washdown_25-001_Full Set_Stamped.pdf	\Vaulted MRV\Facility expansion calculation and evidence\
InProgress_VD_Facilities_Map_v1.pdf	\Vaulted MRV\Facility expansion calculation and evidence\
Kansas site expansion emissions.xlsx	\Vaulted MRV\Facility expansion calculation and evidence\
Vaulted 8_25 Update.xlsx	\Vaulted MRV\Facility expansion calculation and evidence\
AR-09 KOLAR June 2025.pdf	\Vaulted MRV\Vaulted Reports\
Screenshot 2025-08-15 at 9.27.25 AM.png	\Vaulted MRV\Vaulted Reports\
Screenshot 2025-08-15 at 9.27.38 AM.png	\Vaulted MRV\Vaulted Reports\

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Supplier Response	Conclusion / Resolution
1	Clarification	For the biochar moisture content analyses, can you please clarify the sampling approach for collecting samples from the supersacks? It is preferable that the sample analyzed is from a composite made of various samples from the supersack to ensure it is representative	The samples for biochar moisture content are a composite of material randomly taken from different parts of randomly chosen supersacks filled during a production period.	Complete. Thank you for the clarification and using appropriate methods.
2	FAR	The PDD should be updated to reflect the new biomass source and associated information	The new biomass source is included in the PDD expansion currently under validation (see Appendix 3 Charm Biomass Feedstock Information -- Arkansas Mountain	Complete.
3	FAR	The PDD should be updated to reflect GHG Statement calculation changes as noted in the referenced briefs	No problem, I can update Section F of V1.10 with any changes not already captured	To be Reviewed at next verification. Accpetable.
4	FAR	Please plan to provide updated documentation on community engagement and stakeholder input activities by 12/31/2025	Will do -- Working with our Community Engagement Manager on this item	Review for future verifications
5	FAR	Please plan to provide updated documentation on grievance mechanism updates and any complaints/grievances received and responses (or confirmation of no issues) by 12/31/2025	Will do -- Working with our Community Engagement Manager on this item	Review for future verifications
6	FAR	Please ensure all waste disposal related emissions for future disposal of wastes which have been produced during thios period are accounted for in future reporting periods as required. Note: Charm has properly accounted for all disposals so far and has all systems in place to do so going forward.	Absolutely -- We will continue to account for completed disposals as soon as they are completed and, if applicable, documentation is provided to quantify mass, confirm disposal method, etc.	Review for future verifications
7	Clarification	In Waste Disposal Emissions Calc Sheet, for the large disposals (Rows 3-7), the container type is indicated as Tanker Truck for one and Plastic tote for all others. Is this correct? Also, lfor Row 3, Tanker Truck is indicated, but Transportation type in Column P is Tractor Trailer / Box Truck. All other similar rows indicate 'Trailer'. Finally - the only emission factor used was for flatbed / non-tanker truck. For Row 3 a Tanker Truck is indicated as container type (and 18 containers). We assume this is an error. Can you please double check and note any required changes?	The tanker truck designation in Row 3 is an error -- All waste disposals were disposed of using flatbed trailers. As noted, because the tanker truck emission factors were used to quantify emissions, this does not impact Net CDR for the reporting period, but is understandably confusing	Complete. Please upload corrected file

8	Recommendation	Please include in the VVB Guidance document which removal(s) in Certify include the reporting period process, site, emissions (all of the deductions). They are spread across three removals this period, and it was difficult to locate certain emissions.	Understood -- Future removals through Certify/Mangrove should be much clearer in this respect. The Ledger software will automatically assign emissions to removals by date until it hits one that is larger than the remaining gross CDR for a given removal, then it will move on to the next one. As a result, emissions can be spread across multiple removals when there are some larger site emisisons, as was the case during this period	Complete. Thank you for attention to this.
9	Clarification	For Site Emission 590 Ajs Generic Non-Diesel Emission, the value is listed in Certify as 18.933 ton CO2e. However, the file attached only provides info on 200 lbs of clay oil dri usage and a total CO2e of 30 kg. This is the same for site emission 593, 591, and 592. Pplease provide the correct evidence in Certify. Note that it appears these four entries of 18933 are divided into 4 allotments (SE 590, 591, 592, 592) over three separate removals to account for the 75.73 MT of CO2 emissions from the railcar waste disposal nondiesel emissions. Please confirm this is the case. In the future, please document in the VVB guidance or elsewhere how these emissions are allocated across removals. It makes it difficult to match evidence against values in Certify if this is not clear.	I can see what happened here -- You are right, the 4 allotments all map to the 75.73 MT CO2e from the railcar disposal nondiesel. The issue is that the way Ledger offsets site emissions against removals, if the site emission is bigger than the outstanding gross removal, it stops allocating. In this case, the emission was larger than any gross removal, so there was no way to allocate it, so we broke it into pieces. This was a "hot fix" while troubleshooting and I completely forgot to add to guidance; apologies for the confusion	Complete. Thank you for the clarification.
10	Clarification	Could you please provide additional information on the derivation of the emission factors for waste oil incineration and landfilling?	The emission factors and evidence to support were included in the December-January Reporting Period docs package. I will send over copies for easy reference	Complete. Information previously provided and validated.
11	Omission	Please provide the bills of lading to validate the Mass shipped for the rail car waste disposal	The mass shipped for railcar waste disposal is documented by the BOLs that AECN provided for the truckloads that were transferred to the railcar. This is the most conservative estimate as it does not account for any loss of mass during the disposal process. This maximizes the estimated emissions as the emission factors for transportation and disposal are scaled by mass. BOL numbers for each railcar are noted at the top of the calc sheet for each railcar. Please note, GATX 023958 and 023975 were never filled, so there are no BOLs or disposal emissions listed for those	Complete. Thank you for the clarification. Conservative approach used and acceptable.

			railcars. Bills of lading are provided in a separate folder that I will send over	
12	Clarification	Can you please point to where the embodied emissions associated with Vaulted and the Vaulted upgrades are included on Certify?	This appears to be a miss -- We updated the CI on the Ledger side to refledct the updated value, but there is no corresponding emission in Certify. After investigating the issue, this appears to be the case in general. We reconstructed the emissions impact by manually scaling the mass of injectate sequestered in each Reporting Period by the appropriate Vaulted embodied emissions CI. The result shows that this discrepancy is clearly sub-material; total Vaulted embodied emissions accounted for .008% of reporting period removals at the low end and .041% at the high end, totalling 482.7 kgs from project initiation in March of 2024 through 7/31/25. While this is clearly non-material, we do want to remedy it to ensure high quality reporting. We will update the next Reporting Period with a 482.7 kg site emission in order to net out all outstanding emissions. Moving forward, our transition to Mangrove/Certify will ensure this is not a repeat occurrence	Complete. To be reviewed at next verification. Approach is acceptable for addressing accurate reporting. Error is not material.
13	Non-Material Misstatement	On the Miniforge Biomass Transport Allocation Spreadsheet, the old emission factors appear to be used on the New Biomas Allocation Tracker tab. This only impacts the emissions by 0.04 tons total and is not material, but shouldbe corrected.	Noted and corrected	Complete. Error is not material.
14	Error / Clarification	In addition, the tare weight used on the NEW Delivery Log tab refers to a tare weight for a 1995 Peterbilt (Tare Scale Ticket tab). When the cited delivery vehicle is a 2007 Peterbilt. This could have potential impact on emissions as the tare weights for prior delivery vehicles are on the order of 35k-36k lbs, while the tare weight for the new delivery vehicle is 45k lbs. Please clarify if this is correct and if a tare weight for the actual delivery vehicle is available.	This was a typo -- I have confirmed with our logistics team that the tare weight corresponds to the 2007 Peterbilt with Cummins motor and Reliance trailer, which is what was used for the deliveries. The emissions calculations are accurate, but the labeling of the tare weight is not and will be revised.	Complete. Thank you for the clarification and correction.
15	Clarification	On MRV Ops notes for Batch 300035, Salt Remaining in Truck is -785 lbs. Can you please explaion why this is a negative value?	Yes -- The negative value shows a net decrease in the quantity of salt left in the truck between batches. This occurs occasionally, when the dip-stick measurement method shows less salt in the tanker trailers than the preceding measurement without any between-injection cleaning. TA new batch of injectate flushes salt residue that was left in the truck from previous injections downhole. The quantity assessed is 70% of	Complete. Thank you for the clarification

			the measured total, in line with the Salt Accounting update brief, to ensure a conservative estimate of salt downhole	
16	Clarification	The files uploaded to Certify for emissions 5088-5091 (rmv_1K2NAMC2D1S0AN45) only provide the mass, not the carbon intensity. The calculations appear correct, but the uploaded evidence isn't complete. All evidence provided in Google drive.	I am not sure this is the case -- When I open the removals in Certify, the Carbon Intensity evidence shows that the CI source document is the MRV Summary for Production Period 6, which shows the calculations behind the CI. That is then scaled by the mass of bio-oil associated with the origin	Complete. Supplier is correct.
17	Recommendation / Note	Brine TOC reports were field for 6/25, 7/7, and 7/28. Technically, these are required to be completed every two weeks in the reporting period (first half of June missing and 3 weeks between samples in July). However, because no injections occurred until June 26, missing June samples is acceptable, And the timing between samples in July does not impac storage operations, so is acceptable, We recommend trying to ensure the schedue is adhered to whenever possible though.	Noted -- As we transition to our own well asset and are able to bring reporting functions in-house, we look forward to smoother and more consistent well monitoring reporting in general	Complete. Monitor in future verifications.
18	Nonconformity - Omission	The Reporting for the storage site is lacking for the reporting period. The following issues were identified: - the July Monthly Report has not been provided and no evidence is provided that it was submitted on time (August 28) - The Q2 Quarterly Report was not submitted on time (due July 28, but extension granted to Aug 28) and is not provided - The Q1 Quarterly Report was submitted late (April 2 instead of March 28) - A quarterly Groundwater Monitoring Report has not been provided for Q2 - The June Monthly Report was submitted late (an extension was granted) This reporting lapse causes significant concerns regarding permit compliance currently and in the future.	We share that concern -- See above response	Complete. Isometric has approved a deviation for this reporting period only. Future periods may not proceed with verification finalization until all reports are provided. (Slack Message Tom Sellers August 15, 2025)

19	FAR	For gas monitoring required by the permit, can you please provide a log of gas monitor results such that we can review and verify that no gas monitor levels have exceeded permit levels which require action? Please provide in next reporting period.	We will request these records from Vaulted	Review in next verification period
20	FAR	Please provide the following reports no later than the next reporting period end: - the July Monthly KOLAR Report - The Q2 Quarterly Report - The Q2 quarterly Groundwater Monitoring Report	We are currently waiting on Vaulted to provide these documents	Review in next verification period
21	Clarification	The July 28 sample of displaced brine was much higher than the June and July 7 samples (3290 mg/l vs ~600). Is there any explanation or concerns regarding this?	Yes -- This was discussed with our principal scientist once received.	Monitor and review in next verification period
22	Forward Action Request (Prior)	350 is aware that Vaulted is completing some equipment and facility upgrades. If applicable to Charm injection processes, the embodied emissions should be included in Charm LCA.	Understood, will confirm with Vaulted prior to the next reporting	Complete - Vaulted updates and equipment additions (May 2025) added to embodied emissions calculatiosn in this reporting period
23	Forward Action Request (Prior)	Non-pyrolysis electricity consumption (G&A, Eng) is based on records from May 2024. Please evaluate current usage for non-pyrolysis activity, as the changes in operations at the facility have been significant and usage may be much different. NEW Comment: Despite stating this would be addressed in this reporting eriod, teh Production period calculations still rely on the May 2024 allocation. Please ensure that this is addressed in the next reporting period.	This has actually been addressed, I accidentally neglected to include it in the reporting package. I worked withour HR team to confirm that non-production headcount (including non-production Ops, finance, facilitiesand R+D engineering) in CO has grown considerably, from 9 in 2024 to 19 in 2025. It is reasonable to assume that non-production electricity consumption generally scales linearly with headcount, so continuing to use the 2024 estimate inherently underestimates the quantity of electricity consumed	Please complete and update by end of next reporting period

			for non-production activities. This constitutes a conservative assumption, since less electricity allocated to non-production activities translates to more electricity allocated to production and netted against the carbon stored in pyrolysis products used in removals	
24	Forward Action Request (Prior)	Lack of data obtained for salt deliveries as noted in Charms Verifier Guidance document provides less certainty in associated emissions and is an indicator of potential data tracking and management issues. Please work with suppliers to ensure proper weigh tickets / BOL are provided for each delivery to ensure accuracy. It is noted that the approach taken to fill data gaps appears conservative and overall does not materially effect the calculations of neet removals. However, it does not instill confidence.	We will continue working with the team on site to better manage delivery drivers and improve scaling consistency	Charm has improved tracking significantly and was only missing one delivery ticket. Acceptable, but continue to monitor to ensure full implementation and consistency.

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	**Criteria Met?**	**Evidence / Notes** *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Guy Ingram-Hardwick

Carbon Removal Verification Engineer, 350Solutions

EDUCATION:

MEng Materials Science and Engineering, Loughborough University, UK - 2022

EXPERIENCE SUMMARY:

Guy Ingram-Hardwick is a Materials Engineer with experience in process engineering, MRV protocol development, experimental design and life cycle analysis (LCA). At 350 Solutions, Guys efforts center on validation and verification of varied carbon removal pathways. Guy began his experience in the carbon markets directly after graduating working to develop an LCA model for Brilliant Planet, a marine based carbon removal company growing, processing and burying microalgae for carbon sequestration. Once completing the LCA model, Guy managed the third-party verification and co-authored a paper currently awaiting publication documenting the findings. Guy also co-authored the MRV methodology for Brilliant Planet before managing its adoption with carbon market registries and developing the relevant PDD documentation. Guy's Master's thesis in degradation mechanisms for biodegradable polymers resulted in another publication but also provided him experience in experimental design and execution which lead to his experience at Brilliant Planet in conducting field trials for developing novel technologies associated to the processing and storage of the microalgae, displaying the long-term permanence of the carbon removal system. Prior to his experience at Brilliant Planet, Guy worked as a process engineer at Pirelli's rubber compound manufacturing plant in Burton-on-Trent with a focus on data analytics for driving continuous improvement.

RESEARCH AND PROFESSIONAL EXPERIENCE:

January 2025 – Present: Carbon Removal Verification Engineer, 350Solutions

Verify carbon dioxide removal technologies on behalf of registries and private companies ensuring high quality and meaningful climate impact.

Jul 2022 – Dec 2025: LCA and MRV Associate, Brilliant Planet

Quantified the carbon removal efficiency of the Brilliant Planet system across a variety of engineering designs using LCA. Developed the proprietary MRV methodology and PDD as well as setting up a novel experimental design to display the permanence of the stored carbon.

July 2019 – September 2020: Process engineer, Pirelli

Completed data analytics to drive continuous improvement for increasing efficiency, safety and rubber compound quality.